



DTCR-U-Net: a dual-task consistency and multi-scale adaptive attention framework for pneumonia lesion segmentation

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Abstract

Pneumonia remains a critical public health challenge, ranking as the third leading cause of death globally. Although deep learning has shown promise for segmenting pneumonia lesions in CT images, it still struggles to model long-range dependencies, effectively integrate multi-scale features, and cope with limited annotated data. These challenges hinder the extraction of fine-grained details and global contextual information of image regions, leading to blurred or incomplete segmentations, particularly for complex structures and lesion boundaries. To address these issues, a novel segmentation framework, Dual-Task Consistency regularized U-Net (DTCR-U-Net), is proposed, integrating a dual-task consistency regularization (DTCR) architecture with a multi-scale adaptive attention (MSAA) mechanism. Key innovations include: (1) task-level consistency regularization loss to jointly optimize segmentation and boundary detection tasks, ensuring geometric information alignment; (2) the MSAA mechanism, comprising Channel Contextual Enhancement (CCE) and Multi-Scale Global Spatial (MSGs) modules, to model inter-channel dependencies and cross-scale spatial correlations; (3) a multi-scale skip connection (MsSC) scheme to aggregate multi-level features during decoding; and (4) a semi-supervised learning paradigm to leverage unlabeled samples for enhanced generalizability with limited labeled data. Extensive experiments on the MosMedData+LIDC-IDRI, COVID-19-CT-Seg, and CC-CCII datasets demonstrate the superiority of DTCR-U-Net. On MosMedData+LIDC-IDRI, DTCR-U-Net achieves a Dice of 80.10%, F1 score of 83.76%, Sensitivity of 92.37%, Specificity of 99.87%, and Accuracy of 95.68%, outperforming all state-of-the-art baselines. Consistent improvements are observed on the CC-CCII benchmark (Dice 80.05%, F1 score 83.51%). Furthermore, under a semi-supervised setting on COVID-19-CT-Seg, DTCR-U-Net achieves the lowest Hausdorff Distance (4.947 mm) and the highest Dice score (80.12%). These results demonstrate that DTCR-U-Net effectively tackles the long-range dependency modeling, multi-scale feature integration, and data scarcity, exhibiting strong robustness and promising clinical applicability for pneumonia lesion segmentation in CT images.

Keywords Consistency regularization · Semi-supervised learning · U-Net · Skip connection · Attention mechanism · Pneumonia lesion segmentation

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1 Introduction

The human respiratory system, with the lungs as its central organ, plays a crucial role in maintaining vital gas exchange and overall physiological homeostasis. However, pneumonia, an acute respiratory infection caused by bacterial or viral pathogens, presents as a mild illness in healthy individuals and remains a leading cause of high mortality in vulnerable populations, particularly older adults and children. According to the report of the World Health Organization (WHO), respiratory diseases rank as the third leading cause of death worldwide, with pneumonia alone accounting for approximately 3 million deaths annually [1]. In the United States, pneumonia leads to over 1 million hospitalizations and nearly 500,000 deaths each year [2]. The growing burden of pneumonia, exacerbated by population aging and worsening environmental pollution, underscores the critical need for advanced diagnostic methods and early intervention strategies.

Apparently, the early diagnosis and timely intervention of pneumonia are critical for reducing fatality rates [3]. Current diagnostic paradigms primarily rely on medical imaging, each modality offering specific strengths and limitations. Chest X-ray (CXR), although inexpensive and widely accessible, suffers from limited resolution, often resulting in missed diagnoses of early-stage or mild infections. Furthermore, the overlapping imaging characteristics of bacterial and viral pneumonia complicate differential diagnosis [4]. Computed tomography (CT) scans provide superior resolution and detailed anatomical information, making them particularly valuable for diagnosing complex cases. However, its clinical utility is constrained by factors including radiation exposure, technical complexity, and prohibitive costs, rendering it less feasible for large-scale screening applications [5].

The interpretation of CT findings—especially patterns such as ground-glass opacities—is fundamental for assessing pneumonia severity and guiding clinical decision-making. However, the complex textures and indistinct boundaries of infected regions pose significant challenges for accurately segmenting pneumonia lesions. In addition, the scarcity of high-quality annotated datasets, exacerbated by the labor-intensive nature of medical image labeling and the requisite domain expertise, presents a substantial barrier to the development of robust segmentation algorithms [6]. This data scarcity is particularly pronounced in three-dimensional (3D) medical imaging, where the annotation process is markedly more complex and time-consuming.

Traditional medical image segmentation techniques typically rely on manually designed features and demonstrate limited clinical utility, particularly in handling image noise, artifacts, and complex lesion patterns. The advent of deep

learning has revolutionized medical image analysis, with Convolutional Neural Networks (CNNs) [7] emerging as a cornerstone technology. The evolution from conventional CNNs to Fully Convolutional Networks (FCNs) [8] and subsequently to U-Net [9] has significantly advanced medical image segmentation capabilities.

Recently, a large number of U-Net-based variants have been found in the literature. For instance, the U²-Net [10], featuring a two-stage nested U-shaped design, incorporates a novel Residual U-block (RSU) that efficiently extracts multiscale features while preserving feature map resolution, delivering competitive performance even with training from Scratch. Lan et al. [11] proposed BRAU-Net++, a hybrid network that combines CNN and Transformer architectures. This model utilizes a two-layer routing attention mechanism as a foundational component within its U-shaped encoder-decoder framework, enabling it to capture global semantic information while maintaining computational efficiency effectively. Additionally, it enhances skip connections by incorporating channel-space attention through convolutional operations. MetricUNet [12] integrated metric learning to comprehensively model voxel-to-voxel relationships, prompting the network's intermediate feature maps to form a discriminative embedding space for precise voxel-level classification. This metric learning module is lightweight but versatile, facilitating easy integration into existing architectures without requiring additional learnable parameters or significant modifications. Along similar lines, Saha et al. [51] recently introduced AUNet-MHA, which embeds a Multi-Head Self-Attention (MHA) module within an Attention U-Net framework to enhance lung lesion segmentation.

Concurrently, Semi-supervised learning (SSL) approaches have demonstrated remarkable potential in mitigating the data annotation bottleneck. Notable advancements include the uncertainty-aware pseudo-labeling (UPS) framework [13], graph-guided mean teacher models [14], uncertainty-aware time-integrated (UATE) model [15], shape-aware semi-supervised segmentation approach [16] and consistency regularization techniques, exemplified by FxMatch [17] with its integration of entropy minimization [18, 19] and consistency regularization [20]. These studies highlight that regularization in the loss function can effectively improve prediction consistency for unlabeled data.

Despite these advancements, several critical challenges persist in pneumonia lesion segmentation, which motivates our work. First, existing methods, including advanced attention models like AUNet-MHA, primarily operate in a fully-supervised manner, failing to address the fundamental problem of Limited Utilization of Unlabeled Data in a field constrained by scarce annotations. Second, while task-level consistency regularization has been explored, its application remains nascent, leading to Insufficient Inter-task Synergy,

where the complementary relationship between region and boundary information is not fully exploited. Finally, many architectures still suffer from Inefficient Feature Integration, where conventional skip connections limit the fusion of multi-scale features, hindering the segmentation of small lesions and the precise delineation of complex boundaries.

To overcome these specific challenges, this paper proposes a novel framework, DTCR-U-Net (Dual-Task Consistency Regularized U-Net), designed to enhance segmentation accuracy and boundary delineation. The proposed approach integrates a segmentation task with a boundary detection task within a synergistic optimization framework, reinforced by task-level consistency regularization. By leveraging geometric information from the boundary detection task, the proposed approach integrates a segmentation task with a boundary detection task within a synergistic optimization framework, reinforced by task-level consistency regularization loss ($\mathcal{L}_{DTC}$). Furthermore, we introduce a Multi-Scale Adaptive Attention (MSAA) mechanism, comprising the Channel Contextual Enhancement (CCE) and Multi-Scale Global Spatial (MSGs) modules, which collaboratively model inter-channel dependencies and capture cross-scale spatial correlations for more effective feature fusion. Additionally, a multi-scale skip connection mechanism, integrating residual and dense connections in the decoding stage, enhances cross-scale feature representation, mitigates semantic gaps, and preserves fine-grained details, thereby significantly improving segmentation robustness. By fully utilizing both labeled and unlabeled data, DTCR-U-Net reduces reliance on expensive annotated datasets and provides an efficient, robust semi-supervised solution for pneumonia lesion segmentation. The main contributions of this article can be summarized as follows:

- (1) **A Novel Dual-Task Consistency Regularization (DTCR) Framework:** To tackle the issue of insufficient labeled data in pneumonia CT image segmentation, this article introduces a deep neural network framework based on dual-task consistency learning. An innovative task-level consistency regularization loss is proposed to achieve task-level consistency regularization by jointly optimizing the segmentation and the boundary detection task to ensure spatial coherence. While the segmentation task captures regional information, the boundary detection task accurately localizes lesion boundaries, enabling precise segmentation and boundary refinement of pneumonia lesion regions.
- (2) **Long-Range Dependency Modeling with Multi-Scale Spatial Context:** To address semantic information loss and the challenge of modeling long-range dependencies in pneumonia CT image segmentation, this article proposes the MSGS module. This component incorporates

global contextual information to capture cross-scale spatial dependencies at the decoder stage, improving lesion localization accuracy, particularly for small regions. Additionally, a multi-scale skip connection scheme, integrating residual and dense connections with multi-scale aggregation, bridges semantic gaps by progressively aggregating cross-layer features. This facilitates long-range feature interactions, enhancing the model's ability to perceive complex lesion structures.

- (3) **Efficient Multi-Scale Feature Fusion with Channel Attention:** To enhance the challenge of multi-scale feature fusion, this paper introduces the CCE module, which employs a dot-product self-attention mechanism to model inter-channel correlations. This mechanism dynamically adjusts the weight distribution of feature channels, suppressing redundant information and enhancing key semantic features, thereby achieving efficient multi-scale feature fusion. Moreover, a multi-scale feature embedding strategy integrates features across different scales during the encoding stage, ensuring consistency between global and local semantic information. This significantly enhances feature representation, leading to superior segmentation performance.

The rest of this article is organized as follows. Section 2 briefly reviews the principles of Skip connections in U-Net and the consistent regularization in semi-supervised learning, highlighting existing limitations in practical applications. Section 3 details the architecture and modules of the proposed DTCR-U-Net. Section 4 presents the confirmatory and comparative experiments, along with their results and discussions. Section 5 summarizes the entire work, outlining potential further research directions.

2 Related works

This section first reviews the skip-connection mechanism in U-Net and its variants, then surveys the use of consistency regularization in SSL. We focus on how these techniques are applied in practice and critically analyze their inherent limitations, which in turn motivate the design choices of our method.

2.1 Skip connections in U-Net-based architecture

U-Net is a widely adopted encoder-decoder-based CNN for medical image segmentation. Its end-to-end architecture leverages multi-scale feature fusion via skip connections, which substantially improves the delineation of delicate anatomical structures. These connections play a crucial role in preserving spatial information lost during the encoding

process, facilitating the reconstruction of high-resolution segmentation maps in the decoder. However, the traditional skip connection mechanism has inherent limitations: feature fusion is typically performed only between encoder and decoder layers at the same resolution, while the potential semantic discrepancy between low-level encoder features and high-level decoder representations is largely ignored. To alleviate this issue, a variety of enhanced skip-connection strategies have been proposed, which can be broadly grouped into two categories: (1) dense-connection-based skip enhancements and (2) attention-based skip connections.

2.1.1 Dense connections-based skip enhancement

To mitigate the semantic gap between encoder and decoder features, Zhou et al. [21] introduced the UNet+++, which refines skip connections through a series of densely connected intermediate nodes in the decoder subnetwork, progressively aggregating features across multiple semantic scales. Building on this idea, Qian et al. [22] proposed a hybrid scheme that combines dense and full-scale skip connections, enabling simultaneous capture of fine-grained details and coarse-level semantics and thus improving the flexibility of feature fusion. MultiResUNet [23] further addressed the issue by introducing the ResPath module, leveraging a residual structure to improve cross-scale feature fusion and semantic consistency. Additionally, Taghanaki [24] proposed a lightweight, learnable skip connection mechanism utilizing a selective-attention transmission gate, which dynamically prioritizes discriminative channels and critical spatial regions, simultaneously enhancing segmentation accuracy while reducing memory footprint and network complexity.

2.1.2 Attention-based skip connection designs

Attention mechanisms have also been extensively explored to refine skip connections by enhancing both channel and spatial dependency modeling. FAT-Net [25] incorporates three Feature Adaptive Modules (FAMs) into its encoder–decoder framework to adaptively align the feature distributions between encoder and decoder. In contrast, a memory-efficient decoder aggregates both local and global context by integrating features from parallel dual-stream encoders. Similarly, MSD-Net [26], a multiscale collaborative discriminative network, incorporates a channel attention mechanism comprising a channel attention block, cascaded pyramid convolution block, and residual refinement block to utilize multiscale spatial context and semantically coherent features efficiently. MA-UNet [27] introduced a dual attention mechanism that concurrently models dependencies in

spatial and channel dimensions, establishing explicit correlations between attention responses and feature representations to enhance global contextual reasoning.

2.1.3 Limitations and challenges

Despite the significant advancements achieved through these enhanced skip connection strategies, several challenges remain. First, many methods still struggle to capture long-range semantic dependencies due to the limited effective receptive field of convolutional operators, which constrains their ability to model relations between distant regions. Second, most approaches primarily emphasize channel attention while underexploring spatial relationships across multi-scale features, which restricts their ability to fully exploit semantic disparity information. These limitations become particularly pronounced when segmenting complex lesions or structures whose appearance depends on long-range contextual cues. Consequently, there is a pressing need for more advanced skip-connection designs that can jointly balance spatial and channel attention, strengthen global feature representation, and more effectively integrate information across scales.

2.2 Consistency regularization

2.2.1 Concept of consistency regularization

Consistency regularization is a core paradigm in unsupervised and semi-supervised learning that aims to enhance model generalization and stability by enforcing prediction invariance under perturbations. The central assumption is that a robust model should yield consistent outputs for the same underlying sample, even when the input is corrupted, augmented, or observed from different views. By explicitly constraining such consistency, these methods can effectively exploit unlabeled data and thereby alleviate the reliance on large-scale annotated datasets.

Consistency regularization has shown remarkable effectiveness in computer vision and, in particular, in semi-supervised medical image segmentation. For instance, Sajjadi et al. [28] proposed a stochastic transformation and perturbation regularization method for deep SSL, where the model learns from unlabeled images by minimizing prediction discrepancies across multiple augmented versions of the same sample. Similarly, Tarvainen et al. [29] further refined this concept by introducing the teacher-student consistency framework, in which the student model is trained to align its predictions with those of a more stable teacher model, enabling effective learning from unlabeled data through self-ensembling.

2.2.2 Task-level consistency regularization

Most current consistency regularization approaches focus primarily on data-level perturbation consistency, overlooking the potential of task-level consistency. Task-level consistency regularization emphasizes the synergistic relationship between distinct tasks, allowing for cross-task information sharing that not only enhances individual task performance but also significantly improves the efficiency of leveraging unlabeled data. For example, Li et al. [30] developed a shape-aware semi-supervised segmentation strategy by designing a multi-task deep network that predicts both semantic segmentation and signed distance maps (SDMs) of object surfaces. Similarly, Chen et al. [31] developed a semi-supervised dual-task coherent learning framework for 3D medical image segmentation, in which two branches simultaneously estimate segmentation and SDMs, and a coherence loss is imposed to enforce consistency between the two tasks. These approaches highlight that transitioning from purely data-level perturbation consistency to task-level consistency provides a powerful means to incorporate structural priors and fully leverage unlabeled data in semi-supervised medical image segmentation.

3 Proposed DTCR-U-Net

This section presents the proposed DTCR-U-Net in detail, including the Dual-Task Consistency Regulation (DTCR) network architecture, dual task-level consistency regularization loss ($\mathcal{L}_{DTC}$), multi-scale adaptive attention (MSAA) mechanism, and multi-scale skip connection (MsSC) scheme.

3.1 DTCR network architecture

DTCR network, illustrated in Fig. 1, is a dual-task learning framework specifically designed to address the challenges of limited labeled data and significant semantic gaps in pneumonia lesion segmentation. Traditional medical image segmentation model predominantly rely solely on a pixel-wise segmentation task to delineate infected regions, often neglecting critical structural information such as region boundaries and geometric features, resulting in suboptimal segmentation accuracy and blurred boundary delineation. To overcome this limitation, DTCR innovatively combines the pixel-level segmentation task (Task 1) with a boundary detection task (Task 2), optimizing both in a synergistic manner through shared representation learning. This dual-task collaborative learning approach significantly enhances segmentation precision and boundary localization.

The core concept of DTCR lies in establishing a shared representation space between Task 1 and Task 2, ensuring that the outputs of both tasks are consistently constrained not only at the pixel level but also concerning geometric structure information, particularly the boundaries of the pneumonia lesion regions. This shared representation bridges the semantic gap between the segmentation and boundary detection tasks, enabling the model to integrate pixel-level consistency with geometric structural accuracy. This is particularly critical for accurately modeling the complex boundaries of pneumonia lesion regions.

To ensure consistency between the two tasks, DTCR employs a task-level consistency loss designed to align the model's outputs across tasks at both the pixel and geometric levels. This design enhances the model's ability to

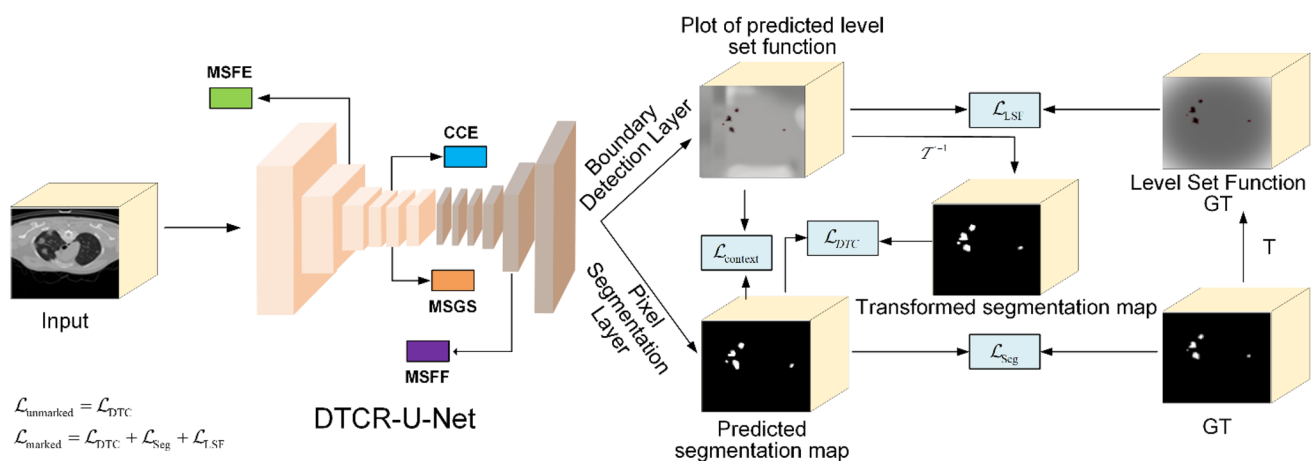


Fig. 1 DTCR network architecture. The framework comprises a pixel segmentation head (Task 1) and a boundary detection regression head (Task 2), utilizing the DTCR-U-Net as the backbone. The boundary detection task predicts the infected region, while the boundary detection task generates a level set function prediction map to capture

fine-grained boundary details. The $\mathcal{L}_{DTC}$ enforces geometric and morphological consistency constraints between the tasks, enabling effective synergistic optimization. This ensures precise segmentation while preserving boundary integrity, enhancing the model's robustness and generalization in pneumonia lesion segmentation

comprehensively understand global and local information, fostering a holistic perspective that significantly improves the accuracy of segmenting infected regions and clarifying boundary details. By jointly optimizing the two tasks within a unified feature space, DTCR attains more stable dual-task learning and delivers superior performance in handling the structural intricacies of pneumonia image segmentation.

3.1.1 Task definition and co-optimization

The DTCR framework employs a dual-task learning paradigm to achieve synergistic optimization between segmentation and boundary detection tasks through shared representations and consistency constraints. Task 1 (pixel-level segmentation) identifies pneumonia-infected regions, capturing global contextual information, whereas Task 2 (boundary detection) delineates precise region boundaries by modeling geometric structures.

Relying on either task in isolation is inherently limited: pure segmentation often struggles to capture intricate and irregular boundaries, while standalone boundary detection lacks sufficient regional context to robustly distinguish lesions from surrounding tissue. To overcome these limitations, DTCR establishes a unified feature space in which regional semantics and geometric cues are jointly encoded and mutually reinforcing, thereby improving both segmentation accuracy and boundary fidelity.

Concretely, Task 1 supplies global region information that guides boundary detection toward infection-related areas, while Task 2 in turn refines the segmentation predictions by Leveraging its strong geometric modeling capability to sharpen boundary representations. To implement this interaction, DTCR leverages the Level Set Function (LSF), a classical mathematical formulation for contour evolution and boundary modeling originally introduced by Osher and Sethian [52]. The LSF provides a flexible representation for capturing object shapes and geometric properties. Furthermore, to promote smooth gradient transitions and enhance boundary consistency, we incorporate a global smoothing regularization term into the LSF-based formulation, which is defined as follows:

$$T(x) = \begin{cases} -\inf_{y \in \partial S} \|x - y\|_2 + \lambda \|\nabla T(x)\|^2, & x \in S_{\text{in}} \\ 0, & x \in \partial S \\ +\inf_{y \in \partial S} \|x - y\|_2 + \lambda \|\nabla T(x)\|^2, & x \in S_{\text{out}} \end{cases} \quad (1)$$

where λ is a regularization coefficient controlling gradient smoothness; $\|\nabla T(x)\|^2$ is a gradient smoothing regularity term; x and y are different pixels in the image, ∂S is the contour (zero-level set) of the target object, S_{in} and S_{out} denote the inner and outer regions of the target object, respectively.

In this way, Task 2 can accurately model the boundary information of the target object.

To enable direct comparability between the outputs of the two tasks, we apply an inverse transformation that maps the boundary-detection prediction back into the segmentation space. Inspired by recent work on task-level consistency, we employ a differentiable sigmoid function to approximate this inverse mapping, making it amenable to end-to-end optimization:

$$\mathcal{T}^{-1}(z) = \sigma(k \cdot z) + R(z) + S(z) = \frac{1}{1 + e^{-kz}} + \lambda_1 \|\nabla z\|^2 + \lambda_2 \|\nabla z\| \quad (2)$$

where $\sigma(k \cdot z) = \frac{1}{1 + e^{-kz}}$ is the Sigmoid function, where k is a constant, $R(z) = \|\nabla z\|^2$ is the gradient smoothing regularization term, used to suppress drastic changes in the predicted values and ensure the continuity and smoothness of the mapping; $S(z) = \|\nabla z\|$ is a Laplacian operator capturing the geometric consistency and spatial properties of the expected values. The regularization terms λ_1 and λ_2 ensure the suppression of abrupt changes in predictions, maintaining smooth and continuous mappings.

By transforming the level set output of Task 2 into the comparable space, the proposed framework establishes inter-task consistency, enabling boundary detection and segmentation results to complement and refine each other, ultimately improving overall segmentation accuracy and boundary clarity. The region-aware capability of Task 1 provides global contextual support for the boundary detection task, while the geometric modeling capability of Task 2, in turn, optimizes the segmentation task's performance in boundary regions.

The shared representation not only enhances the model's performance on labeled data but also enforces output consistency between Task 1 and Task 2 via consistency regularization, thereby improving the utilization efficiency of unlabeled data during training. To further emphasize the interaction of Task 1 and Task 2, we introduce a context-guided loss, $\mathcal{L}_{\text{context}}$, designed to optimize Task 2 by leveraging contextual information from Task 1.

Specifically, the gradient information, $\nabla f_1(x)$ and $\nabla f_2(x)$, capture boundary variations, while a region weighting function $W(x)$ emphasizes critical target areas, guiding Task 2 to focus on relevant regions while suppressing background noise. By minimizing $\mathcal{L}_{\text{context}}$, the gradient information from Task 1 directly informs Task 2, helping it to model boundaries more accurately and maintain consistency in the gradient direction. The formula is as follows:

$$\mathcal{L}_{\text{context}}(x) = \sum_{x \in D} \|\nabla f_2(x) - \nabla(f_1(x) \bullet W(x))\|^2 \quad (3)$$

where $\nabla f_2(x)$ denotes the gradient information of the boundary detection and segmentation task, respectively, and $W(x)$ represents the region weight mapping generated by the segmentation task for emphasizing the impact of the target region.

3.1.2 SSL based on dual-task consistency regularization

The essence of consistency regularization lies in leveraging unlabeled data to enhance the model's generalization ability by imposing consistency constraints on the model's predictions for unlabeled data. Traditional consistency regularization methods primarily focus on data-level perturbations, such as minimizing prediction variance by applying small perturbations to input data, thereby improving the model's robustness to unlabeled data. In contrast, DTCR-U-Net extends this concept to the task level, introducing DTCR scheme that jointly optimizes the segmentation and boundary-detection tasks. This approach enables more effective use of unlabeled data by leveraging complementary information between the two tasks, ultimately improving model robustness and performance.

In DTCR-U-Net, dual-task consistency regularization builds a synergistic optimization framework between the segmentation head (Task 1) and the boundary-detection head (Task 2), particularly under limited labeled data. Unlike conventional semi-supervised methods that treat labeled and unlabeled samples differently at the loss level, DTCR-U-Net enforces inter-task consistency on both labeled and unlabeled data, allowing unlabeled samples to directly drive parameter updates. In this way, the model substantially reduces its dependence on expensive pixel-wise annotations while simultaneously improving segmentation accuracy and boundary delineation.

The dual-task consistency regularization loss ($\mathcal{L}_{\text{DTC}}$) ensures consistency between the two tasks by minimizing the prediction discrepancy between the segmentation task (Task 1) and the boundary detection task (Task 2) on unlabeled data. A key advantage of this approach is its ability to operate on unlabeled data. Since unlabeled data is more readily available than labeled data, this approach substantially reduces the dependence on labeled data. During training, the output of Task 2 (level set function) must be mapped back to the segmentation output space. Direct computation of the exact inverse is infeasible due to the non-differentiable nature of the level-set operator; therefore, we adopt a differentiable approximation based on a sigmoid-like mapping. The $\mathcal{L}_{\text{DTC}}$ is defined as follows:

$$\mathcal{L}_{\text{DTC}}(x) = \mathcal{L}_{\text{main}}(x) + \lambda_1 \mathcal{L}_{\text{grad}}(x) + \lambda_2 \mathcal{L}_{\text{interact}}(x) \quad (4)$$

where λ_1 , λ_2 are balance coefficients regulating the contribution of each term.

The main consistency loss ($\mathcal{L}_{\text{main}}$) is the core part of the whole loss function, used to constrain the global consistency of the outputs of the two tasks. It is defined as:

$$\mathcal{L}_{\text{main}}(x) = \sum_{x_i \in D} \|f_1(x_i) - \mathcal{T}^{-1}(f_2(x_i))\|^2 \quad (5)$$

where $f_1(x_i)$ is the prediction result of the segmentation task, $f_2(x_i)$ is the prediction result of the boundary detection task, $\mathcal{T}^{-1}(f_2(x_i))$ is the result of the output of the boundary detection task by inverse transformation, which is used to align the output space of the segmentation task; $\|\cdot\|^2$ is the Euclidean distance, used to measure the difference between the outputs of the two tasks.

The gradient consistency loss ($\mathcal{L}_{\text{grad}}$) is used to enhance the model's sensitivity to segmentation boundaries, thus improving the segmentation of complex structures and details, i.e.,

$$\mathcal{L}_{\text{grad}}(x) = \sum_{x_i \in D} \|\nabla f_1(x_i) - \nabla \mathcal{T}^{-1}(f_2(x_i))\|^2 \quad (6)$$

where $\nabla f_1(x_i)$ is the gradient of the segmentation result, which captures the change information of the segmentation boundary, $\nabla \mathcal{T}^{-1}(f_2(x_i))$ is the gradient of the boundary checking result after passing the inverse transformation.

Interaction Enhancement Loss ($\mathcal{L}_{\text{interact}}$) is used to further mine the synergistic information between the segmentation task and the boundary detection task, emphasizing the joint effect of the two. The formula is defined as:

$$\mathcal{L}_{\text{interact}}(x) = \sum_{x_i \in D} \|f_1(x_i) \odot f_2(x_i)\| \quad (7)$$

where $f_1(x_i) \odot f_2(x_i)$ is the element-by-point product of the segmentation result and the boundary detection result, capturing the interaction response of the two on the target region.

Assume that D_1 and D_2 represent the labeled and unlabeled datasets, respectively, with the total dataset denoted as $D = D_1 \cup D_2$. For the labeled dataset D_1 , each pixel-level sample pair is represented as (x, y) , where y denotes the ground truth labeling for the segmentation task. For the unlabeled dataset D_2 , only the input data x is available. The semi-supervised mechanism operates as follows: for an unlabeled sample x from the dataset D_2 (denote as x_u for clarity), the model performs a single forward pass. This generates two predictions: $f_1(x_u)$ from the

segmentation head and $f_2(x_u)$ from the boundary head. Since no ground truth y exists for x_u , the consistency loss $\mathcal{L}_{DTC}$ is computed solely based on the discrepancy between these two internal predictions, i.e., it measures the discrepancy between $f_1(x_u)$ and the transformed output $T^{-1}(f_2(x_u))$. In this way, unlabeled samples directly contribute to parameter updates by enforcing that the two task-specific heads learn a coherent and robust shared representation. It is important to emphasize that the unlabeled subset does not reuse any original manual annotations. For images within D2, the ground-truth masks are completely discarded, and the model receives supervision solely through the dual-task consistency loss. The pseudo-labels involved in training are not hidden human labels but are generated automatically by the boundary-regression head, in line with standard task-coherence-based SSL designs.

In DTCR-U-Net, the segmentation task is optimized by jointly learning from both the segmentation task and the boundary detection task, enabling effective co-learning between labeled and unlabeled data. For the labeled dataset D_1 , the supervised loss of the segmentation task $\mathcal{L}_{seg}$ is based on the widely used Dice loss [Ref_Dice], which is particularly effective for handling class imbalance in segmentation tasks. This enhancement strengthens the segmentation predictions' sensitivity to local contextual information. The supervised segmentation loss is defined as follows:

$$\mathcal{L}_{Seg}(x, y) = \sum_{x_i, y_i} 1 - \frac{2 \sum_{x_j \in v_i, v_j \in v_i} w_{i,j} f_1(x_j) y_j}{\sum_{x_j \in v_i} w_{i,j} f_1(x_j) + \sum_{y_j \in v_i} w_{i,j} y_j + \epsilon} \quad (8)$$

where v_i denotes the neighborhood of voxel x_i in the 3D data; $w_{i,j}$ is the weight based on spatial similarity; $f_1(x_j)$ is the prediction result of the segmentation task; y_j is the true label; and ϵ is a small constant to avoid the denominator being zero.

Then we define the supervised loss of the boundary detection task as the loss between the predicted level set function and the transformed ground truth map L . The formula is:

$$\mathcal{L}_{LSF}(x, y) = \sum_{x_i, y_i \in D} \|f_2(x_i) - T(y_i)\|^2 + \beta \cdot \|\nabla f_2(x_i) - \nabla T(y_i)\|^2 \quad (9)$$

where $\|f_2(x_i) - T(y_i)\|^2$ is used to ensure the overall consistency of the boundary detection results with the real boundary; $\|\nabla f_2(x_i) - \nabla T(y_i)\|^2$ strengthens the boundary detection task's ability to capture boundary details through gradient constraints; β is a hyperparameter controlling the weights of the gradient terms.

The final loss integrates both supervised and semi-supervised components, defined as,

$$\mathcal{L}_{total} = \mathcal{L}_{seg} + \mathcal{L}_{LSF} + \lambda_d \mathcal{L}_{DTC}. \quad (10)$$

where $\mathcal{L}_{seg}$ and $\mathcal{L}_{LSF}$ are used only for labeled data, $\mathcal{L}_{DTC}$ is used for both labeled and unlabeled data during training, so that the segmentation task and the detection task can optimize the network in a semi-supervised manner using either labeled or unlabeled data jointly; $\lambda_d(t) = e^{(-5(1-\frac{t}{t_{max}})^2)}$.

3.2 MSAA mechanism

The MSAA mechanism is designed to bridge the semantic gap between the encoder and decoder by jointly optimizing feature representations across both channel and spatial dimensions. As depicted in Fig. 2, the MSAA mechanism comprises two key sub-modules: the Channel Contextual Fusion (CCE) and the Multiscale Global Spatial (MSGs) Modules. These two modules independently optimize multiscale features along the channel and spatial dimensions, as visualized in Fig. 3.

Specifically, the CCE module dynamically reweights channel-wise features to amplify task-relevant semantic information while suppressing redundant or irrelevant activations. This selective enhancement improves feature discrimination, ensuring that only the most informative features are retained. Meanwhile, the MSGs module captures long-range spatial dependencies and models global contextual relationships, thereby reinforcing spatial consistency and emphasizing critical regions.

By integrating these two modules, MSAA provides a comprehensive feature refinement mechanism that strengthens both contextual awareness and spatial coherence. By employing the multi-scale adaptive modeling of the MSAA module, we effectively mitigate the semantic gap between the encoder and decoder. Furthermore, this approach significantly enhances the accuracy and robustness of segmentation tasks, offering robust support for medical image segmentation in complex scenarios.

3.2.1 Multi-scale feature embedding

To effectively capture cross-scale contextual information and enhance feature representation of an original image $X \in \mathbb{R}^{H \times W}$, we perform feature reshaping and tokenization operations on the feature maps $E_i \in \mathbb{R}^{\frac{H}{2^{i-1}} \times \frac{W}{2^{i-1}} \times C_i}$, ($i=1, 2, 3, 4$) extracted at each stage of the encoder to achieve the fusion and expression of cross-scale features. The procedures are as follows:

(1) Channel Alignment: To ensure consistency across different scales, each feature map E_i of each scale is channel-mapped by 1×1 convolution to unify the channel dimension. All features are projected to a fixed dimension $C=128$:

$$E_i' = \text{Conv}_{1 \times 1}(E_i), i = 1, 2, 3, 4. \quad (11)$$

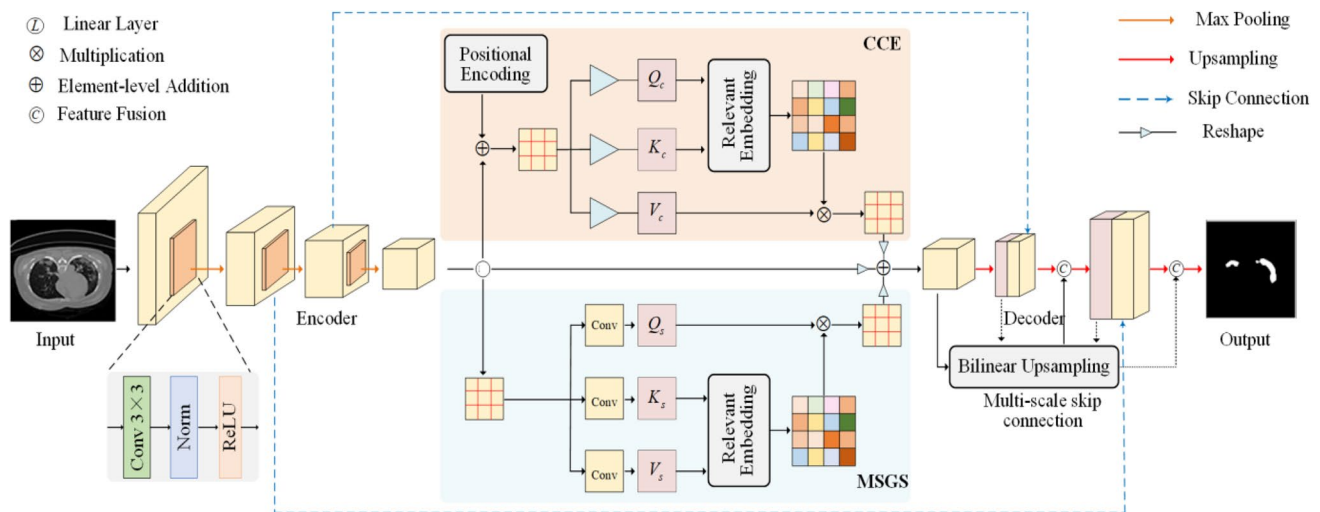


Fig. 2 Schematic diagram of the internal structure of DTCR-U-Net: **a** CCE and MSGS are embedded into the MSAA module, the encoder extracts features through multi-layer convolution, and combines the CCE and MSGS to optimize the feature representations from the channel and spatial dimensions, respectively, to capture cross-scalar

context information; **b** the decoder effectively aggregates multi-level features through the multi-scale skip connection and feature fusion mechanisms to aggregate multi-level features and significantly improve segmentation performance

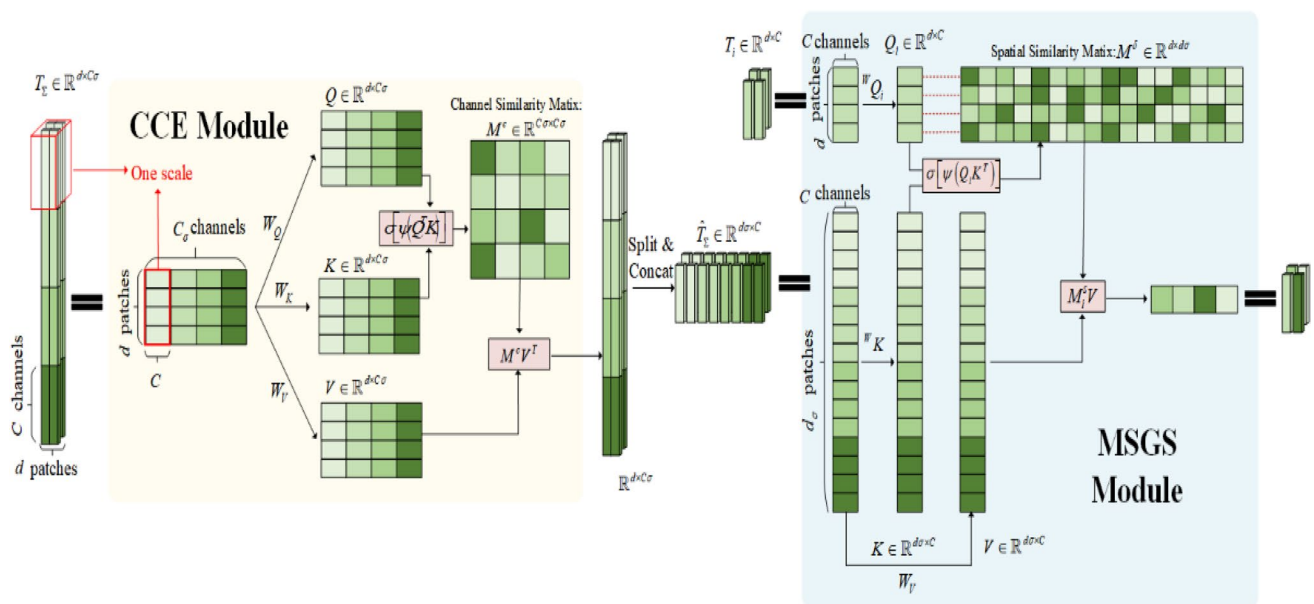


Fig. 3 Visualization of the learning process in the proposed CCE and MSGS modules. These modules integrate channel and spatial attention mechanisms to effectively capture multi-scale representational features from distinct dimensions of global dependencies. Features at varying

scales are distinguished using color coding, providing a clear depiction of the modules' operational mechanisms and their role in learning and optimizing multi-scale features

(2) Feature Reshaping and Tokenization: Then the feature map E_i^c at each scale is reshaped into a marker sequence T_i , where the marker sizes are $P, \frac{P}{2}, \frac{P}{4}, \frac{P}{8}$, and the length of the marker sequence at each scale is $d = \frac{HW}{P^2}$:

$$T_i = \text{Reshape}(\text{Conv}_{1 \times 1}(E_i)), i = 1, 2, 3, 4. \quad (12)$$

where p denotes the window size of the markers and d denotes the length of the sequence.

(3) Multi-scale Feature Fusion: Then, we spliced four different scales of marker sequences T_1, T_2, T_3, T_4 along the channel axis to form a multi-scale fused representation of the markers T_Σ :

$$T_{\Sigma} = \text{concat}_c([T_1, T_2, T_3, T_4]), T_{\Sigma} \in \mathbb{R}^{C_{\sigma} \times d}, C_{\sigma} = 4C. \quad (13)$$

3.2.2 CCE module

Different channels inherently encode scale-specific semantic features. The effective fusion of these channel-wise representations is crucial for enhancing segmentation performance, as raw multi-scale feature embeddings may introduce redundant or imbalanced information. To address this, the CCE module is designed to model inter-channel dependencies, selectively emphasizing informative features while suppressing irrelevant or redundant components.

As illustrated in Fig. 3, the CCE module employs a channel self-attention mechanism to enhance feature representation along the channel dimension. This mechanism enables the model to learn global semantic dependencies by capturing relationships among channels within the multi-scale feature representation T_{Σ} . Specifically, the CCE module first models the relationships between channels by learning queries (Q_c) keys (K_c) and values (V_c) via three independent linear transformations:

$$Q_c = W_Q T_{\Sigma}, K_c = W_K T_{\Sigma}, V_c = W_V T_{\Sigma} \quad (14)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{C_{\Sigma} \times C_{\Sigma}}$ are the learned mapping matrices, d is the sequence length, and $C_{\sigma} = 4C$ is the full-scale channel dimension.

To capture inter-channel dependencies, we compute a similarity matrix via scaled dot-product attention, followed by normalization using the Softmax function:

$$M_c = \text{Softmax} \left(\frac{Q_c \cdot K_c^T}{\sqrt{C_{\Sigma}}} \right), M_c \in \mathbb{R}^{C_{\Sigma} \times C_{\Sigma}}. \quad (15)$$

The channel attention matrix M_c is then used to weigh V_c and integrate the original channel features using a residual connection to enable feature fusion and optimization across different stages. To further enhance information representation among channels, we introduce a channel balance factor γ and a bias term β . The channel features F_{CCE} are obtained by adaptive calibration with parameters learned through Global Channel Pooling (GCP):

$$F_{CCE} = \gamma \cdot (M_c \cdot V_c + T_{\Sigma}) + \beta \quad (16)$$

where $\gamma = \text{Sigmoid} \left(\text{FC} \left(\text{GAP} \left(M_c \cdot V_c + T_{\Sigma} \right) \right) \right)$, $\beta = \text{FC} \left(\text{GAP} \left(M_c \cdot V_c + T_{\Sigma} \right) \right)$.

Here, Global Average Pooling (GAP) aggregates channel-wise information, while a fully connected (FC) layer learns the balance factor. The Sigmoid function constrains the learned weights within $[0, 1]$, ensuring stable adaptation.

Unlike conventional self-attention mechanisms that operate along the spatial dimension, the CCE module focuses exclusively on the channel axis, enabling it to effectively highlight discriminative channel-wise features. Additionally, instance normalization is applied to stabilize training by normalizing the similarity matrix, facilitating smooth gradient propagation. By dynamically adjusting inter-channel weights, the CCE module enhances the model's ability to capture lesion-specific information, filter out irrelevant features, and improve overall feature representation.

3.2.3 MSGS module

The spatial attention mechanism is critical for capturing long-range dependencies in feature maps, particularly in challenging segmentation scenarios, such as small target regions, blurred boundaries, and complex anatomical structures. Traditional convolution operations, constrained by their limited receptive field, struggle to capture such dependencies effectively. To address this limitation and enhance the representation of multi-scale spatial information in medical image segmentation, we propose MSGS module, which integrates global and local spatial information through a multi-scale spatial attention mechanism. This module emphasizes lesion regions and boundary details, thereby improving the model's ability to perceive and segment complex target regions.

To implement the spatial dimensionality of the attention mechanism, we first apply a linear transformation to the enhanced channel features F_{CCE} to generate queries Q_s , keys K_s , and values V_s matrices:

$$Q_s = W_Q * F_{CCE}, K_s = W_K * F_{CCE}, V_s = W_V * F_{CCE}. \quad (17)$$

where $W_Q, W_K, W_V \in \mathbb{R}^{1 \times 1 \times C}$ are 1×1 convolution kernels, and $*$ denotes the convolution operation.

The transformed feature maps are then flattened into a two-dimensional matrix representation:

$$Q_s, K_s \in \mathbb{R}^{C \times HW}. \quad (18)$$

Then, the spatial similarity matrix M_s is computed using the Scaled Dot-Product Attention:

$$M_s = \text{Softmax} \left(\frac{Q_s^T \cdot K_s}{\sqrt{C}} \right), M_s \in \mathbb{R}^{HW \times HW}. \quad (19)$$

where Q_s^T denotes the transpose of the query matrix, $\frac{1}{\sqrt{C}}$ is the scaling factor to prevent gradient explosion, and the Softmax operation is used to normalize the similarity matrix M_s .

The spatial attention matrix M_s is weighted and summed over the value matrix V_s to obtain the enhanced spatial feature map:

$$F_{MSGS} = \text{Reshape}(M_s \cdot \text{Reshape}(V_s)) + F_{CCE}. \quad (20)$$

where $\text{Reshape}(V_s) \in \mathbb{R}^{C \times HW}$ converts V_s into a two-dimensional form, $M_s \cdot V_s$ denotes weighting the spatial dimensions to capture the global spatial dependencies, F_{CCE} denotes the original features directly transmitted by the residual paths to avoid information loss.

To further enhance the multi-scale perception of spatial features, we introduce convolution kernel pairs of different sizes (3×3 , 5×5 , 7×7) to convolve F_{MSGS} , and fuse multi-scale features:

$$F_{multi-MSGS} = \sum_{s=1}^S \alpha_s \cdot (W_k^{(s)} * F_{MSGS}). \quad (21)$$

where $W_k^{(s)}$ denotes the s -th scale convolutional kernel, α_s is the learnable weight that indicates the importance of each feature, S denotes the number of multi-scale convolutional kernels, and the output $F_{multi-MSGS} \in \mathbb{R}^{C \times H \times W}$ contains global dependencies with multi-scale spatial information.

By employing multiscale convolutional fusion, the MSGS module overcomes the challenge of insufficient small-target and boundary representation at a single scale. The combination of self-attention and residual learning mitigates vanishing gradients, stabilizing training and enhancing segmentation accuracy. This design significantly improves the model's ability to recognize fine details, accurately delineate lesion boundaries, and achieve robust segmentation performance in medical imaging applications.

3.2.4 Attention embedding fusion

To fully leverage the contextual information and spatial relationships obtained by the previous modules, we introduce an Attention Embedding Fusion (AEF) mechanism to achieve complementary optimization between channel and spatial attention. This fusion process enhances key feature representations, ensuring improved segmentation performance by integrating both channel-wise and spatial information in a unified framework.

Since the output of the CCE module (F_{CCE}) and MSGS module (F_{MSGS}) may differ in channel and spatial dimensions, we first employ a 1×1 convolution to align their channel dimensions, facilitating effective fusion:

$$F'_{CCE} = \text{Conv}_{1 \times 1}(F_{CCE}), F'_{CCE} \in \mathbb{R}^{H \times W \times C} \quad (22)$$

$$F'_{MSGS} = \text{Conv}_{1 \times 1}(F_{MSGS}), F'_{MSGS} \in \mathbb{R}^{H \times W \times C} \quad (23)$$

To emphasize informative features, we extract channel-level attention weights A_c from F'_{CCE} using GAP followed by a Softmax activation:

$$A_c = \text{Softmax}(\text{GAP}(F'_{CCE})) \quad (24)$$

where GAP denotes the global average pooling operation for aggregating each channel over the spatial dimension $H \times W$.

Similarly, we compute spatial-level attention weights A_s from F'_{MSGS} :

$$A_s = \text{Softmax}(\text{Conv}_{1 \times 1}(F'_{MSGS})). \quad (25)$$

The final fused representation F_{AEF} is obtained by element-wise weighted fusion of the channel and spatial attention-enhanced feature:

$$F_{AEF} = A_c \odot F'_{CCE} + A_s \odot F'_{MSGS} \quad (26)$$

Where $\odot$ denotes an element-by-element multiplication operation, enabling selective enhancement of important features while suppressing less relevant information. To ensure stable convergence, we initialize A_c, A_s with small random variations drawn from a Gaussian distribution: $A_s, A_c \sim N(0, \sigma^2)$, where σ is a small positive constant, ensuring minor initial differences in attention weights that gradually adapt during training to highlight the most discriminative features. This fusion mechanism further optimizes feature representation by enforcing semantic consistency across channel and spatial dimensions, thereby enhancing the robustness and accuracy of medical image segmentation.

3.3 MsSC scheme

Recent works [25, 32] have underscored the importance of multi-scale feature fusion in capturing both global and local contextual information. In particular, the introduced MsSC strategies facilitate effective feature integration across different semantic scales through a series of transitional operations, including upsampling, concatenation, and convolution. Three distinct types of connections are illustrated in Fig. 4.

(1) Cascade Connection: Cascade connections concatenate multiple feature maps along the channel dimension, enabling direct feature aggregation from different network layers. This approach preserves essential information across

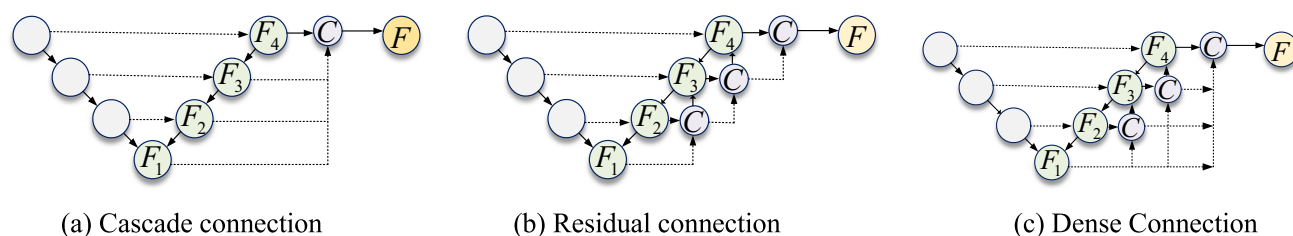


Fig. 4 Comparison of cascade connections with residual or dense connections. **a** Cascade connection: explicitly integrates features from multiple layers, and the number of output channels increases as the number of layers increases; **b** Residual connection: adds the inputs of a

layer directly to the outputs, solving the gradient vanishing problem of deep networks; **c** Dense connection: splices the outputs of all previous layers into the inputs of the current layer, significantly reducing the gradient vanishing problem

layers and is particularly advantageous for skip connections between encoders and decoders. The operation can be expressed as:

$$F_{\text{concat}} = \text{Concat}([F_1, F_2]) \in \mathbb{R}^{H \times W \times (C_1 + C_2)} \quad (27)$$

where $\text{Concat}(\cdot)$ represents the splicing operation along the channel dimension.

(2) Residual Connection: Residual connections, originally introduced in ResNet [33], which introduced “short-cut connections” to address issues such as vanishing and exploding gradients in deep networks. By learning residual mappings between inputs and outputs, this mechanism ensures efficient gradient flow and facilitates stable training. Mathematically, it is defined as:

$$F_{\text{residual}} = F_{\text{low}} + \mathcal{H}(F_{\text{high}}) \quad (28)$$

where F_{low} and F_{high} respectively represent the low-level and high-level feature maps, $\mathcal{H}(\cdot)$ represents the transforms extracted by convolution or attention mechanisms.

(3) Dense Connection: Dense connection, as proposed in DenseNet [34], differs from cascade connection by connecting the output feature maps of each layer to the output feature maps of all preceding layers. This forms a multilevel cascade of features, enabling effective feature reuse, reducing redundant computations, and enhancing the network’s feature representation capability. The process can be formulated as:

$$F_{\text{dense}} = \text{Concat}([F_{\text{low}}, F_{\text{mid}}, F_{\text{high}}]) \quad (29)$$

where F_{mid} represents the characterization map of the middle layer.

The proposed DTCR-U-Net integrates two advanced multi-scale skip connectivity mechanisms—residual connectivity and dense connectivity—to effectively guide the decoder’s upsampling process. Initially, high-level semantic features are fused with low-level spatial features via residual

connectivity to ensure efficient information flow. Subsequently, dense connectivity aggregates features across multiple scales, followed by convolutional operations for further refinement. To enhance contextual awareness, multi-scale convolutional kernels operate at varying receptive fields before the final fused features are weighted adaptively to generate the skip-connected output feature map F_{out} :

$$F_{\text{out}} = \sum_i^N \alpha_i \cdot \mathcal{F}(\text{Concat}([F_{\text{low}}, F_{\text{mid}}, F_{\text{high}}])) + F_{\text{low}} + \mathcal{H}(F_{\text{high}}) \quad (30)$$

Unlike conventional approaches that rely solely on single-pass cascade connections, the stepwise growth enabled by residual and dense connections facilitates the progressive aggregation of multi-scale decoder features. This approach enhances feature representation by combining hierarchical information across different semantic levels. Consequently, DTCR-U-Net effectively captures both global structures and fine-grained details, ensuring robust segmentation of lesion regions at varying scales. Moreover, this strategy mitigates the loss of spatial details commonly induced by excessive upsampling while addressing challenges such as gradient vanishing and overfitting, ultimately leading to improved model performance and robustness.

3.4 Comparative analysis with existing dual-task and attention-driven architectures

Recent advances in dual-task learning and attention-based architectures have substantially improved the performance of medical image segmentation. Nevertheless, most existing approaches still face notable limitations in three key aspects: effective multi-scale feature fusion, structural consistency across tasks, and efficient exploitation of unlabeled data. To better position the advantages of the proposed DTCR-U-Net within the broader research landscape, this section provides a comparative analysis against representative dual-task frameworks, attention-based models, and conventional fully supervised segmentation networks.

3.4.1 Comparisons with existing dual-task learning frameworks

Dual-task segmentation models typically couple semantic segmentation with auxiliary tasks such as signed distance map (SDM) regression or multi-branch feature learning, aiming to jointly encode region-level and boundary-level cues. While these methods generally enhance boundary localization, they can exhibit limited robustness when confronting ambiguous lesion shapes, highly irregular boundaries, or large-scale anatomical variations. Their performance is also susceptible to annotation noise, and the additional task branches often introduce non-trivial computational overhead.

In contrast, DTCR-U-Net adopts a reversible-mapping-based dual-task design that explicitly enforces consistency between region and boundary predictions. By integrating region-level, boundary-level, and task-level constraints into a unified optimization framework, DTCR-U-Net yields more stable multi-task learning dynamics and stronger segmentation robustness, particularly in anatomically complex or low-contrast regions where traditional dual-task schemes tend to deteriorate.

3.4.2 Comparisons with attention-mechanism-based segmentation models

Attention-based architectures, such as AUNet-MHA [51], MA-UNet [27], and FAT-Net [25], refine feature selection and model long-range dependencies by incorporating channel-wise and/or spatial attention mechanisms. However, these designs often emphasize high-level semantic features and provide only limited modeling of cross-scale spatial correlations, which can hinder the precise delineation of small lesions and fine anatomical boundaries. In addition, many of these models focus primarily on single-task optimization and do not fully exploit potential synergies between complementary tasks.

DTCR-U-Net addresses these limitations through the MSAA mechanism, which decouples channel and spatial modeling using the CCE and MSGS modules. Combined with the MsSC strategy, MSAA facilitates richer multi-scale feature interaction and improves sensitivity to subtle lesion structures, resulting in more accurate and robust segmentation performance.

Distinct from existing attention-based and single-task approaches, DTCR-U-Net further incorporates a DTCR framework that enables task-level co-optimization by enforcing consistency constraints between the segmentation and boundary-detection branches. The proposed DTCR framework jointly captures region-level semantics

and boundary geometry, thereby overcoming the inherent limitations of purely single-task designs and significantly improving the utilization efficiency of unlabeled data in a semi-supervised setting. Moreover, unlike many prior methods that rely on multiple inference paths or teacher–student architectures, DTCR-U-Net achieves task-level regularization within a single forward pass, which effectively reduces computational overhead and enhances its practicality for real-world medical imaging applications.

4 Experimental validation and result discussions

This section details the experimental procedure and results as well as the corresponding performance analysis and discussions.

4.1 Dataset description and preprocessing

To comprehensively evaluate the proposed method, we conduct experiments on four publicly available chest CT datasets.

MosMedData [35]. The MosMedData, provided by medical institutions in Moscow, Russia, contains a total of 1,110 CT volumes, of which only 50 volumes were annotated by radiologists. The annotations are binary masks: pneumonia-infected regions are represented by the value 1 (white), while all remaining pixels are represented by 0 (black). While a subset of images include manual annotations of infected regions, the majority of CT scans are accompanied only by categorical labels without pixel-level segmentation annotations.

LIDC-IDRI [36]. This dataset is provided by the National Cancer Institute (NCI) and the Lung Image Database Consortium (LIDC), consisting of 2,633 CT image sets from 1,018 patients. Detailed annotations are available for some nodule regions, which were labeled by multiple radiologists. However, most lung regions are not annotated, and the labeled lesion areas are primarily focused on lung nodules. In some images, the overall lesion region information is missing.

In this experiment, a hybrid dataset comprising the MosMedData and LIDC-IDRI datasets was utilized. First, all CT volumes with in-plane resolution different from 512×512 are resampled to a uniform resolution. Then, for each volume, slices containing annotated lesions are identified, and for each such slice we retain the slice itself together with its two adjacent slices above and below. The resulting dataset is partitioned into training, validation, and test subsets with a 6:2:2 ratio to support fair training and evaluation.

Semi-supervised protocol on MosMedData+LIDC-IDRI: In our semi-supervised setting, a subset of the training scans from the hybrid MosMedData+LIDC-IDRI dataset is intentionally treated as unlabeled: their ground-truth segmentation masks are completely removed during training. Although these scans originate from datasets that contain manual annotations, no annotation information is used for them once they are assigned to the unlabeled pool. Instead, their supervision signal is derived solely from the boundary-detection branch via the proposed dual-task consistency regularization, rather than from human-provided labels. This design ensures that our semi-supervised learning protocol strictly adheres to standard SSL principles, whereby unlabeled images contribute only through model-generated signals.

COVID-19-CT-Seg [37]. The COVID-19-CT-Seg includes 349 CT images with clinical manifestations of COVID-19 from 216 patients. It features three segmentation labels, all annotated by junior radiologists and validated by two senior radiologists. Only certain lesion regions in the CT images are manually labeled, while other regions remain unlabeled. For this study, 80 CT images were randomly selected as the training set, of which 40% are labeled, and 60% are unlabeled. An additional 20 CT images were selected as the test set, all of which are labeled.

CC-CCII [53]. The CC-CCII dataset is a large-scale chest CT benchmark collaboratively collected by several Chinese medical institutions to support research in pneumonia diagnosis. It comprises 40,830 CT series from 4,154 patients, covering a wide spectrum of clinical conditions, including viral pneumonia, common pneumonia, and healthy controls. A key characteristic of CC-CCII, and one that makes it particularly relevant for our semi-supervised framework, is that many scans are provided without fine-grained pixel-level lesion annotations, closely reflecting the annotation scarcity commonly encountered in real-world clinical practice.

To improve model training efficiency and optimize computational complexity while ensuring consistency and diversity in the input data, the following preprocessing steps were applied:

(1) Image Conversion and Normalization: The original images were converted to grayscale, removing redundant color channel information to reduce memory usage and computational overhead. The pixel values were normalized and scaled from $[0, 255]$ to $[0, 1]$, accelerating gradient computation and optimization during model training. Additionally, contrast enhancement of lesion regions in the lungs was performed using histogram equalization to make lesion features more distinct. Contrast adjustment was further applied to improve the separability of lesion regions across different images.

(2) Downsampling and Data Augmentation: The high-resolution images (512×512) were downsampled to 256×256 to reduce computational complexity. Key slices from each CT image group were extracted, and the diversity of labeled data was expanded using augmentation techniques such as flipping, cropping, and rotation. Unlabeled data distributions were uniformly normalized to maintain consistency across the dataset.

The preprocessed images are then fed into DTCR-U-Net for training. All models are trained for 80 epochs with the AdamW optimizer, using an initial learning rate of 0.001 and a cosine annealing schedule to gradually decay the learning rate and promote smooth convergence. The batch size is set to 32. During training, unlabeled samples from the target domain are included in the loss computation through the dual-task consistency terms, and the model checkpoint with the lowest validation loss is selected as the final model.

All experiments are implemented in PyTorch (version 1.9.1) and conducted on a workstation equipped with an NVIDIA RTX 4090 GPU (24 GB VRAM) and 30 GB of system memory. Under this configuration, training DTCR-U-Net for 80 epochs requires approximately 18 h.

To evaluate the feasibility of deploying DTCR-U-Net in real clinical environments, we additionally measured the inference latency, GPU memory occupancy, and TensorRT-accelerated performance on commonly-used radiology GPUs. For $512 \times 512 \times N$ CT volumes, DTCR-U-Net achieves an average inference speed of 78 ms per slice (2.45 s per 32-slice volume) on an NVIDIA RTX-3080, and 41 ms per slice (1.28 s per volume) on an RTX-4090. The peak GPU memory consumption is 3.2 GB and 2.7 GB, respectively, well below the typical 16-GB budget available in clinical workstations. With TensorRT FP16 optimization, the model reaches 109 ms (RTX 3080) and 56 ms (RTX 4090) per forward pass, which is consistent with the expected latency for a 53.74-GFLOP network. These results demonstrate that DTCR-U-Net is computationally efficient and practically deployable within routine clinical radiology workflows.

4.2 Dataset diversity and model generalization capacity

To alleviate potential concerns regarding demographic or acquisition homogeneity, we emphasize that our study does not rely solely on MosMedData and the LIDC-IDRI subset. In addition to these datasets, we incorporate two large-scale and highly heterogeneous benchmarks—COVID-19-CT-Seg and CC-CCII—as well as unlabeled multi-center CT scans within our semi-supervised learning framework.

The CC-CCII dataset contains 40,830 CT series from 4,154 patients collected across multiple hospitals, covering a broad spectrum of demographics, scanning devices, disease severities, reconstruction kernels, and slice thickness values (including acquisitions up to 5 mm). COVID-19-CT-Seg further enriches variability by providing multi-center annotated scans from patients of different ages and clinical stages.

This combination ensures substantial diversity in image appearance, acquisition protocols, and pathological patterns, effectively mitigating the limitations of using MosMedData and LIDC-IDRI alone. Furthermore, the DTCR-U-Net framework leverages heterogeneous unlabeled data through semi-supervised learning, significantly improving robustness under domain shift. The strong performance observed on CC-CCII validates that the proposed model generalizes well to real-world heterogeneous clinical environments.

4.3 Evaluation metrics

Five evaluation metrics are used for performance evaluation: Dice coefficient, Hausdorff distance (HD), Sensitivity (SE), Specificity (SP), and Accuracy. To assess the segmentation accuracy, the binary segmentation map output from the model was compared pixel by pixel with the expert-labeled ground truth map. Based on the comparison results, pixels were classified into four categories: TP (True Positive), FN (False Negative), TN (True Negative), and FP (False Positive).

Sensitivity (SE) represents the true positive rate, which measures the probability that an actual positive case is correctly predicted as positive. Specificity (SP) represents the true negative rate, which measures the probability that an actual negative case is correctly predicted as negative. The formulas for SE and SP are as follows:

$$SE = \frac{TP}{TP + FN} \quad (31)$$

$$SP = \frac{TN}{TN + FP} \quad (32)$$

The Dice coefficient is a commonly used metric for assessing segmentation model performance, specifically measuring the overlap between predicted outputs and ground truth regions. It ranges from 0 to 1, with values closer to 1 indicating superior segmentation accuracy. The formula is given as follows:

$$\text{Dice} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (33)$$

The Hausdorff distance (HD) is a metric used to quantify the distance between two boundary sets, making it particularly effective for assessing the worst-case deviation between the predicted boundary and the true boundary in segmentation tasks. A smaller HD value indicates that the predicted boundary is closer to the true boundary. The formula is as follows:

$$HD(P, T) = \max \{ \sup_{p \in P} \inf_{t \in T} d(p, t), \sup_{t \in T} \inf_{p \in P} d(t, p) \} \quad (34)$$

where $\sup_{p \in P} \inf_{t \in T} d(p, t)$ denotes the maximum value of the shortest distance from a point p in the predicted boundary to the real boundary T , $\sup_{t \in T} \inf_{p \in P} d(t, p)$ denotes the maximum value of the shortest distance from a point t in the real boundary to the predicted boundary P .

Accuracy, a key metric in classification tasks, evaluates the overall correctness of a model's predictions across all samples, where a value approaching 1 reflects higher performance. The formula is as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (35)$$

The F1 score balances a model's precision and recall, offering a comprehensive performance measure. Precision denotes the proportion of correctly predicted positive instances among all positive predictions, while recall reflects the fraction of actual positives correctly identified. By combining these metrics, the F1 score provides an overall assessment of the model's effectiveness. The formula is as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (36)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (37)$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (38)$$

4.4 Comparative experiments and result analysis

To comprehensively assess the effectiveness of the proposed DTCR-U-Net, we conducted comparative experiments against state-of-the-art pneumonia segmentation models on the hybrid MosMedData and LIDC-IDRI datasets. Additionally, it was evaluated against recently proposed semi-supervised learning methods on the COVID-19-CT-Seg dataset.

4.4.1 Comparisons with state-of-the-art pneumonia segmentation models

(1) Comparative Experiments

We first benchmark DTCR-U-Net against several representative pneumonia segmentation models. The quantitative results on the MosMedData+LIDC-IDRI hybrid dataset are summarized in Table 1, and qualitative visualizations are presented in Fig. 5. These comparisons enable a systematic evaluation of DTCR-U-Net in terms of overall segmentation accuracy and visual delineation quality of pneumonia lesions.

BCS-Net [38] introduces an Attention-Guided Global Context (AGGC) module to selectively propagate critical encoder features to the decoder, thereby improving information flow. A Semantic Guidance (SG) unit is further employed to merge multi-scale high-level features and refine decoder outputs through SG mapping.

CdcSegNet [39] proposes a continuous dilated convolution (CDC) module to enlarge the receptive field and capture richer contextual information, coupled with an adaptive multicore pooling (AMP) module designed to alleviate information loss during downsampling.

NormNet [40] utilized a voxel-level anomaly modeling approach to distinguish between healthy tissues and COVID-19-affected regions, generating synthetic “lesions” through random shapes, noise generation, and image filtering.

DC-UNet [41] redesigned the encoder-decoder architecture with an efficient CNN architecture and replaced traditional skip connections with a residual module.

SSA-Net [42] introduced a Spatial Self-Attention Network within an encoder-decoder framework, expanding the receptive field and enhancing self-learning while proposing a semi-supervised iterative segmentation framework for multi-class infection segmentation using pseudo-labels to expand the training dataset.

2.5-D DPE-Net [43] employed a dual-path encoder network with multi-scale dynamic weights (MsDWDSM) to fuse spatial and inter-slice features, improving lesion localization and segmentation accuracy.

PDAtt-Unet [44] utilized an input pyramid to maintain spatial awareness across encoder layers and designed attention gate-based decoders for infection and lung segmentation, alongside a hybrid loss function to address boundary pixel segmentation ambiguities.

As shown in Table 1, DTCR-U-Net consistently outperforms existing models across all metrics. Notably, it achieves an SP (specificity) of 99.87%, significantly higher

Table 1 Comparisons with existing methods on the hybrid MosMedData and LIDC-IDRI dataset

Methods	SP(%)	SE(%)	Accuracy(%)	Dice(%)	F1(%)	Parameters(M)	FLOPs(G)
BCS-Net [38]	98.93	83.28	92.94	75.95	76.75	44.82	87.98
CdcSegNet [39]	99.21	84.31	92.43	74.98	80.66	31.64	183.15
Norm Net [40]	99.29	88.75	94.05	76.93	75.09	40.48	170.23
DC-UNet [41]	99.05	86.63	94.76	62.74	78.84	51.79	95.78
SSA-Net [42]	99.47	90.83	93.95	77.64	79.85	47.53	58.92
2.5-DDPE-Net [43]	99.38	91.24	95.21	75.78	81.65	3.05	52.26
PDAtt-Unet [44]	99.25	90.67	94.89	78.83	80.74	43.18	82.74
DTCR-U-Net	99.87	92.37	95.68	80.10	83.76	18.65	53.74

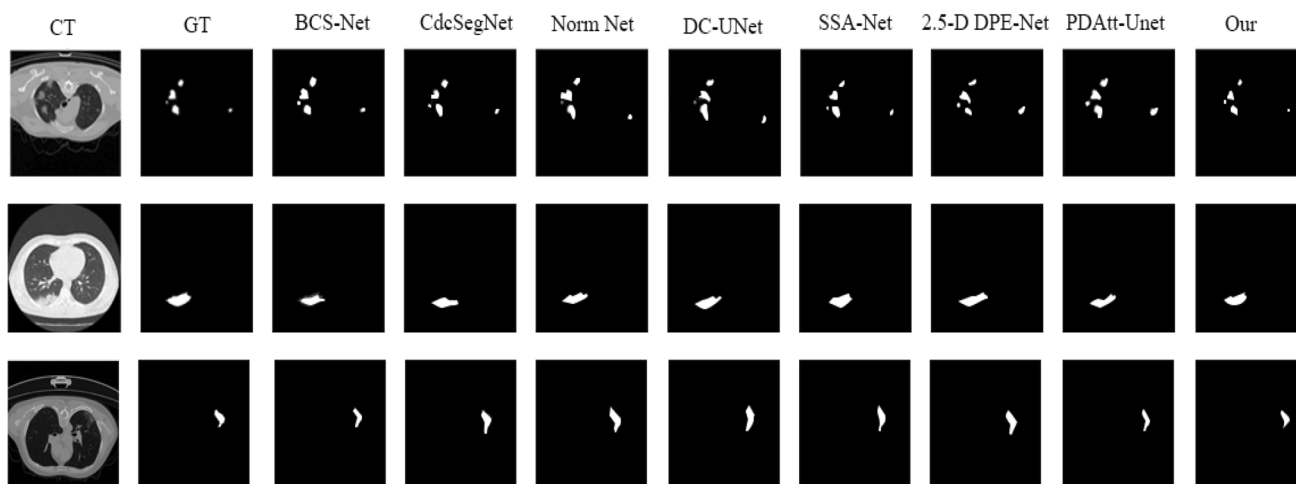


Fig. 5 Visualization of comparative segmentation results

than these comparative methods, demonstrating its ability to minimize false positives and accurately identify healthy regions. This strong background suppression capability reflects the effectiveness of the CCE module within the MSAA mechanism, which optimally models feature channels to reduce misclassification of background regions.

In the SE (sensitivity) metric, DTCR-U-Net attains an excellent score of 92.37%, showcasing its ability to capture critical lesion regions and reduce missed detections. This robustness, particularly in small lesion regions and fuzzy boundaries, is attributed to the Task-Level Consistency Regularization (TLCR) incorporated into the model. TLCR fosters synergy between the segmentation and boundary detection tasks, striking a balance between capturing lesion boundary details (SE) and maintaining background specificity (SP).

DTCR-U-Net also achieves a leading Accuracy score of 95.68%, demonstrating its adaptability and reliability in overall segmentation tasks. The multi-scale skip connection mechanism effectively aggregates features from different decoding stages, enhancing segmentation accuracy, particularly for complex lesion morphologies.

In terms of the Dice coefficient and F1 score, DTCR-U-Net achieves 80.10% and 83.76%, respectively, underscoring its superiority in fine segmentation tasks. These metrics reflect the model's ability to balance precision and recall while accurately capturing geometric properties and detailed lesion information. This success is largely due to the MSGS module, which models cross-scale spatial dependencies, enhancing the fusion of global and local features.

(2) Patient-level robustness and statistical significance

Beyond the aggregate metrics in Table 1, we further assess model stability and statistical reliability by computing the patient-level mean and standard deviation (Mean $\pm$ Std) of the Dice coefficient across all test cases. As summarized in Table 2, DTCR-U-Net achieves a Dice score of $80.10\% \pm 2.41$, while PDAtt-UNet and SSA-Net obtain $78.83\% \pm 2.56$ and $77.64\% \pm 2.73$, respectively. These results indicate that DTCR-U-Net not only performs better on average but also maintains more consistent segmentation quality across patients.

To examine whether these improvements are statistically significant, we conduct a Wilcoxon signed-rank test on patient-level paired Dice scores. Considering seven pairwise comparisons among the models, we adopt a Bonferroni-corrected significance level of ($p < 0.0071$). As shown in Table 2, DTCR-U-Net yields statistically significant gains over all competing methods, with all p -values falling below the corrected threshold. This confirms that the observed performance advantages of DTCR-U-Net are both numerically superior and statistically robust.

Table 2 Patient-level dice mean $\pm$ Std and Wilcoxon signed-rank test results ($n=30$)

Methods	Dice (%)	p -value vs DTCR-U-Net	Significant ($p < 0.0071$)
BCS-Net [38]	75.95 $\pm$ 3.01	<0.001	Yes
CdcSegNet [39]	74.98 $\pm$ 3.15	<0.001	Yes
Norm Net [40]	76.93 $\pm$ 2.84	0.0021	Yes
SSA-Net [42]	77.64 $\pm$ 2.73	0.0045	Yes
2.5-DDPE-Net [43]	75.78 $\pm$ 2.95	<0.001	Yes
PDAtt-UNet [44]	78.83 $\pm$ 2.56	0.0032	Yes
DTCR-U-Net	80.10 $\pm$ 2.41	-	-

Overall, these results demonstrate that DTCR-U-Net provides an advanced and robust solution for pneumonia segmentation, outperforming current state-of-the-art models in terms of both accuracy and stability. In particular, its ability to handle challenging regions—such as fuzzy boundaries and small or scattered lesions—highlights the effectiveness of the proposed MSAA mechanism, MsSC scheme, and dual-task consistency regularization in real-world medical imaging scenarios.

4.4.2 Validation on the CC-CCII benchmark for generalizability

To rigorously assess the generalizability of the proposed framework, we further evaluate DTCR-U-Net on the CC-CCII dataset. This setting enables a direct, fair comparison with several state-of-the-art segmentation methods under a standardized evaluation protocol. The quantitative results are summarized in Table 3.

The results on CC-CCII are consistent with those obtained on the hybrid MosMedData+LIDC-IDRI dataset and provide strong additional evidence of the effectiveness and generalizability of DTCR-U-Net. From Table 3, several key observations can be drawn:

- (1) **State-of-the-Art Segmentation Accuracy:** DTCR-U-Net establishes a new state-of-the-art on this benchmark, achieving the highest performance across all core metrics. It obtains a Dice score of 80.05% and an F1 score of 83.51%, corresponding to an absolute Dice improvement of approximately 2.4 percentage points over the next-best method, SSA-Net. This consistent advantage highlights the strong representational capacity of the proposed MSAA module. By jointly capturing global contextual information and multi-scale spatial dependencies, MSAA enables more accurate localization and segmentation of complex pneumonia lesions.
- (2) **Superior Boundary Delineation and Lesion Identification.** Beyond overall accuracy, DTCR-U-Net exhibits clear benefits in handling challenging structural details.

Table 3 Comparisons with existing methods on CC-CCII datasets

Methods	SP(%)	SE(%)	Accuracy(%)	Dice(%)	F1(%)
BCS-Net [38]	98.79	83.28	92.94	75.95	76.75
CdcSegNet [39]	99.21	84.31	92.43	74.98	80.66
Norm Net [40]	99.29	88.75	94.05	76.93	75.09
SSA-Net [42]	99.47	90.83	93.95	77.64	79.85
PDAtt-Unet [44]	99.38	91.24	95.21	75.78	81.65
DTCR-U-Net	99.85	92.15	95.52	80.05	83.51

It achieves the highest sensitivity (SE) of 92.15%, indicating an excellent ability to correctly identify lesion regions and reduce missed detections—an essential property for reliable clinical decision-making. This improved perception of ambiguous and fine-grained boundaries is largely attributable to the dual-task consistency regularization (DTCR) framework, in which the dedicated boundary-detection branch provides explicit geometric guidance to the segmentation branch. Importantly, this high sensitivity is achieved while simultaneously maintaining the highest specificity (SP) of 99.85%, demonstrating that the model effectively suppresses background noise and avoids spurious false-positive activations.

In summary, the consistently strong performance of DTCR-U-Net on this independent and challenging benchmark provides compelling evidence of its generalizability. The results confirm that the core architectural and algorithmic innovations—MSAA, MsSC, and dual-task consistency regularization—transfer effectively across datasets, yielding a model that excels in both overall segmentation accuracy and precise delineation of intricate lesion boundaries.

4.4.3 Comparisons with semi-supervised segmentation methods

To specifically evaluate the effectiveness of our proposed dual-task consistency regularization in a data-constrained environment, we further conducted experiments on the COVID-19-CT-Seg dataset under a semi-supervised setting (40% labeled, 60% unlabeled data). This section therefore focuses on a direct and fair comparison with other

state-of-the-art SSL methods. Six representative semi-supervised segmentation approaches are selected as baselines. Quantitative results are reported in Table 4, and representative qualitative comparisons are shown in Fig. 6.

MC-Net+ [45] proposes a shared encoder with multiple slightly different decoders to encourage consistent, low-entropy predictions in complex regions, thereby improving the exploitation of unlabeled data. It introduces a mutual-consistency scheme that jointly leverages prediction consistency and entropy minimization to regularize the learning process.

URPC [46] pioneers the use of pyramid consistency within a single network for semi-supervised segmentation and introduces a forward-pass-based uncertainty estimation mechanism. By measuring and integrating multi-scale prediction differences into the pyramid consistency framework, the method enables more reliable learning from unlabeled samples.

RCPS [47] presents a semi-supervised segmentation strategy that generates two differently augmented views of each input and adopts a triple-branch learning scheme. It incorporates a two-way voxel-level contrastive loss and a confidence-aware negative sampling strategy, extending contrastive learning into voxel space with bi-directional computations and thereby enhancing semantic separability in the feature space.

FSSL [48] addresses multi-site medical image segmentation by combining prototype estimation under uncertainty with a joint pseudo-labeling strategy. This design produces more reliable pseudo-labels across heterogeneous client sites, enabling effective utilization of unlabeled data in federated or distributed settings.

Table 4 Comparison with existing semi-supervised segmentation methods on the COVID-19-CT-Seg dataset

Methods	HD (mm)	F1(%)	Accuracy(%)	Dice (%)
MC-Net+ [45]	5.8727	75.45	92.78	79.63
URPC [46]	6.4845	78.42	93.26	71.31
RCPS [47]	5.9263	74.93	94.73	75.72
FSSL [48]	5.5402	81.24	93.95	78.63
FreeMatch [49]	5.4628	79.58	93.52	78.16
SCO-SSL [50]	5.9283	80.37	93.95	77.63
DTCN U-Net	4.9471	83.76	95.68	80.12

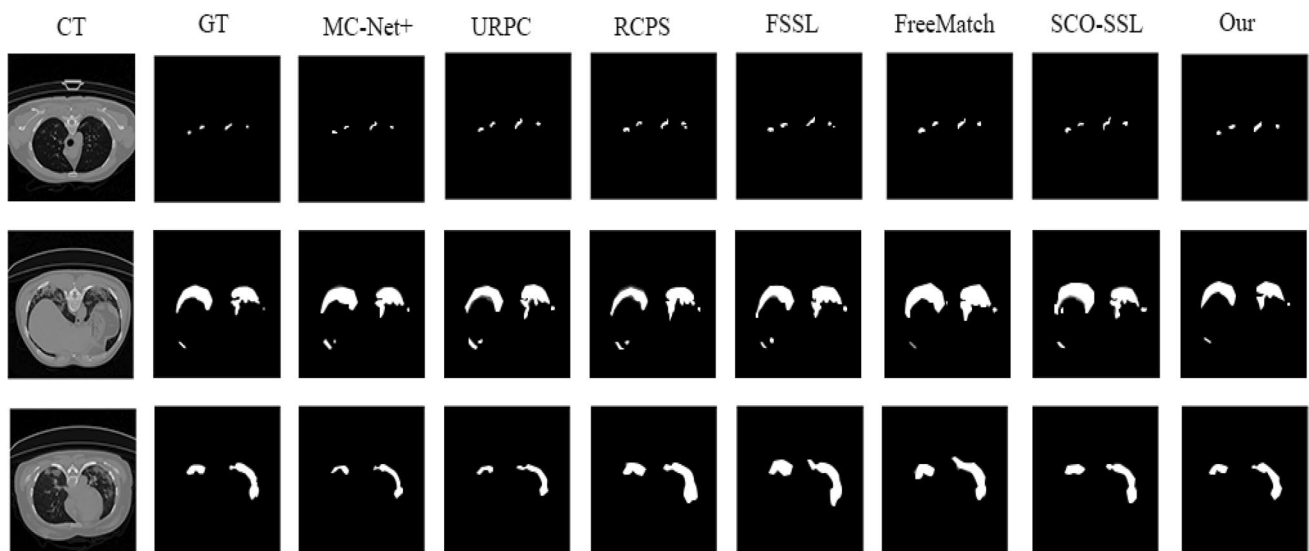


Fig. 6 Comparison of segmentation results. The first column on the left presents the CT images, while the second column displays the corresponding ground truth label maps. The remaining columns

showcase segmentation results from our proposed model compared to those generated by state-of-the-art SSL models

FreeMatch [49] adaptively adjusts confidence thresholds according to the model's learning state. Its Self-Adaptive Thresholding (SAT) mechanism eliminates the need for manually tuned thresholds, while Self-Adaptive Class Fairness Regularization (SAF) encourages balanced, diverse predictions across classes, improving the robustness of semi-supervised learning.

SCO-SSL [50] introduces Shadow Augmentation (Shadow-AUG) and Shadow Loss (Shadow-DROP) for shadow-robust segmentation. Shadow-AUG augments training images with synthetic shadow artifacts to improve robustness to appearance variations, whereas Shadow-DROP exploits neighboring unshadowed pixels to infer boundaries in shadowed regions, thereby enhancing segmentation accuracy in challenging areas.

As shown in Table 4, DTCR-U-Net achieves the best performance across all evaluation metrics. It obtains the lowest Hausdorff Distance (HD) of 4.9471 mm, indicating superior boundary accuracy and a stronger ability to capture complex lesion contours. The substantially reduced HD compared with competing SSL methods highlights the advantage of explicitly modeling lesion geometry through the boundary-detection branch and enforcing dual-task consistency.

Additionally, DTCR-U-Net achieves an F1 score of 83.76%, an Accuracy of 95.68%, and a Dice coefficient of 80.12%, all of which surpass the corresponding scores of the baseline methods. These results demonstrate that the proposed framework can simultaneously capture global contextual information and perform fine-grained delineation of local structures. The low HD reflects its strength in precise boundary prediction, while the high F1 score evidences a favorable balance between precision and recall. The leading

Dice coefficient further confirms that DTCR-U-Net yields stable and accurate segmentation of lesion regions, even under limited supervision.

Overall, the results on COVID-19-CT-Seg validate that the proposed dual-task consistency regularization is highly effective in semi-supervised settings: DTCR-U-Net not only exploits unlabeled data more efficiently than existing SSL approaches, but also provides more accurate and reliable delineation of pneumonia lesions in terms of both region coverage and boundary precision.

4.5 Ablation experiments

To validate the effectiveness of the proposed MSAA module and the dual-task consistency loss ($\mathcal{L}_{DTC}$), multiple sets of ablation experiments are conducted. These experiments validate the contribution of these modules and different parts of the loss function to the model performance improvement by gradually introducing them and further analyzing the individual and synergistic effects of each module.

We use the standard U-Net network without the addition of the MSAA module and $\mathcal{L}_{DTC}$ as the baseline model (Baseline). By gradually introducing the MSAA module and $\mathcal{L}_{DTC}$, we design the following experimental configurations: Method 1 introduces the MSAA module (including both CCE and MSGS) in the encoder part of the Baseline model to verify the role of the MSAA module in multiscale feature fusion; Method 2 introduces the loss function of $\mathcal{L}_{DTC}$ in the Baseline model, which is used to verify the role of the bi-task consistency loss in the enhance the contribution of segmentation and boundary detection co-optimization; Method 3 simultaneously introduces the

Table 5 Ablation studies of the proposed MSAA+MsSC architecture and the LDTC semi-supervised framework on MosMedData dataset

Methods	Module			Metrics (%)			
	MSAA	$\mathcal{L}_{DTC}$	MsSC	Dice (%)	HD (mm)	Accuracy (%)	F1(%)
Baseline	–	–	–	74.54	8.1890	90.23	77.22
Method 1	✓	–	–	78.62	7.8723	93.26	81.43
Method 2	✓	✓	–	77.55	6.7892	92.94	81.92
DTCR-U-Net	✓	✓	✓	80.12	4.9471	95.68	83.76

Table 6 Validation of the CCE and MSGS modules on MosMedData dataset

Methods	Module		Metrics (%)			
	CCE	MSGS	Dice (%)	HD (mm)	Accuracy (%)	F1 (%)
Baseline	–	–	74.54	8.1890	90.23	77.22
DTCR-U-Net-A	✓	–	76.82	6.8285	92.21	78.74
DTCR-U-Net-B	–	✓	77.24	6.6287	92.67	79.32
DTCR-U-Net	✓	✓	80.12	4.9471	95.68	83.76

MSAA module, $\mathcal{L}_{DTC}$ and MsSC scheme (DTCR-U-Net) to construct a complete model to validate the enhancement effect of the combination of the two on the overall segmentation performance.

In addition, to further explore the independent contributions of each component within the MSAA module, we design the following experiments: DTCR-U-Net-A introduces only the CCE module in the baseline model, used to validate the enhancement effect of the channel dimensional modeling on the segmentation performance; DTCR-U-Net-B introduces only the MSGS module in the baseline model used to validate the effect of spatial dimension attention mechanism on target region enhancement. These experiments were conducted on the MosMedData dataset, and the experimental results are shown in Tables 5 and 6.

First, to isolate and evaluate the effectiveness of our $\mathcal{L}_{DTC}$ semi-supervised framework, one must compare Method 2 with the Baseline. Both settings use the same standard U-Net architecture, but Method 2 leverages unlabeled data via our proposed $\mathcal{L}_{DTC}$ loss. The significant improvement from 74.54% (Baseline) to 77.55% (Method 2) in Dice score unequivocally validates that our task-level consistency regularization is highly effective.

Second, comparing Method 1 (78.62% Dice) with the Baseline (74.54%) demonstrates the power of our advanced architecture (MSAA and MsSC) alone, even in a fully-supervised setting.

Finally, the most important finding is the powerful synergistic effect revealed by the full DTCR-U-Net model (80.12% Dice). This result, which significantly outperforms both Method 1 and Method 2, proves that our two main contributions are mutually reinforcing: the advanced architecture provides superior features for the consistency framework to regularize, while the consistency framework enables the architecture to be trained more effectively with limited data. Therefore, contrary to a potential misreading

of the table, the $\mathcal{L}_{DTC}$ is not only beneficial but is a critical component driving the overall success of our framework.

According to the results in Table 6, the Dice coefficient and F1 score of DTCR-U-Net-A improved to 76.82% and 78.74%, respectively. This demonstrates that the CCE module effectively captures global semantic information through channel dependency modeling, significantly enhancing the feature representation capability and providing a strong foundation for improving overall segmentation performance. Similarly, the results for DTCR-U-Net-B indicate that the MSGS module plays a pivotal role in highlighting target regions and capturing cross-scale spatial dependencies. The Dice coefficient and F1 score reached 77.24% and 79.32%, respectively, verifying the effectiveness of the MSGS module in enhancing feature representation, emphasizing critical regions, and improving the ability to capture small lesions.

4.6 Discussion on hyperparameter selection

The formulation of our loss function includes several hyperparameters, such as λ_1, λ_2 for the consistency loss $\mathcal{L}_{DTC}$, and β for the supervised boundary loss $\mathcal{L}_{LSF}$. The selection of these parameters was guided by a combination of empirical experimentation and insights from related literature.

We performed a grid search on a small subset of the validation data to determine the optimal values used in our final experiments (e.g., $\lambda_1=1.0, \lambda_2=1.0, \beta=0.5$). During this process, we observed that the model's performance was relatively stable across a reasonable range of these parameters (e.g., λ_1 from 0.1 to 2.0). This suggests that our framework is not critically sensitive to minor variations in these coefficients, which is a desirable property for robust performance.

The coefficients λ_1 and λ_2 for the consistency loss were set to a similar order of magnitude to ensure a balanced contribution from the main, gradient, and interaction consistency terms. The β parameter for the supervised gradient loss was

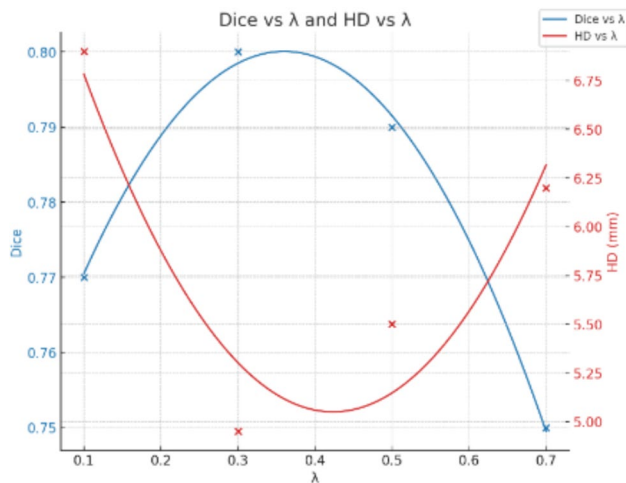


Fig. 7 Relationship between Dice and λ (blue curve) and the relationship between Hausdorff distance and λ (red curve)

set to a slightly smaller value to prioritize the main LSF regression task while still benefiting from the gradient-level supervision. While an exhaustive search across all hyperparameters was computationally prohibitive, our empirical results confirm that the chosen values provide a stable and effective balance between the different learning objectives of our framework.

To validate that the choice of the regularization coefficient λ in Eq. (6) is not dataset-specific, we conducted a sensitivity study in which λ was varied in $\{0.1, 0.3, 0.5, 0.7\}$. For each

setting, segmentation accuracy (Dice) and boundary-quality metrics (HD and ASSD) were computed on the MosMed-Data + LIDC-IDRI dataset.

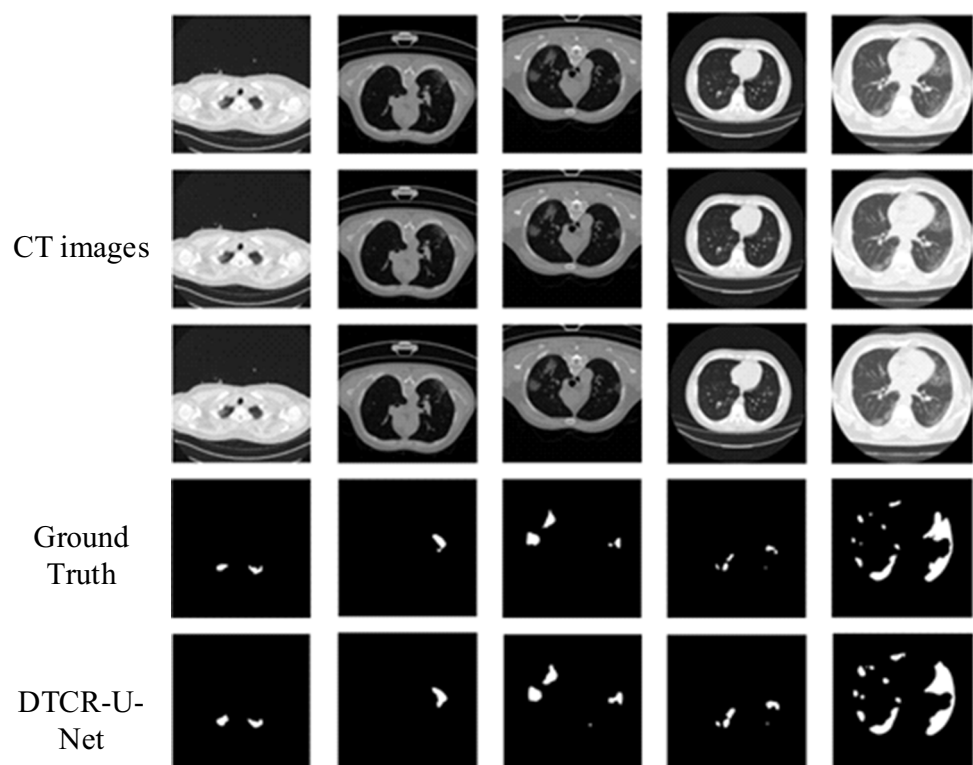
As shown in Fig. 7, $\lambda = 0.3$ yields the optimal Dice score (80.10%) and the lowest HD (4.95 mm). Both overly small λ values (insufficient boundary smoothing) and overly large λ values (excessive smoothing) negatively impact both segmentation accuracy and boundary alignment. These findings verify that $\lambda = 0.3$ achieves a robust balance between region-level and boundary-level consistency, and is not dependent on dataset-specific characteristics.

4.7 Challenging case discussions

To gain deeper insights into the performance of the proposed network, we analyzed several challenging cases where segmentation results were suboptimal, as illustrated in Fig. 8. The following observations were made:

- (1) Boundary Ambiguity and Segmentation Inconsistencies.** The first situation is that complex boundary regions are usually blurred in feature representation, making it challenging to distinguish the demarcation between the background and the lesion. This often results in over-segmentation or under-segmentation. To address this, the geometric modeling capability of the boundary detection task is leveraged to enhance the segmentation task's perception of boundary regions.

Fig. 8 Visualization of poorly segmented segmentation results, the network may generate poorly segmented images when the image contains complex boundary-blurring, breakpoints, voids, or complex textures



Additionally, gradient consistency constraints are incorporated into the loss function to improve sensitivity to boundary details.

- (2) **Internal Lesion Cavities and Small-Scale Feature Loss.** The second situation is that the cavity region inside the lesion may not be adequately learned by the network due to low contrast or noise in CT images. This causes the network to predict the lesion region as a continuous whole. Furthermore, the network may lose small-scale features in the high-level features it generates, failing to accurately preserve lesion details. To resolve this, task-level consistency regularization is employed to strengthen inter-task co-optimization. The segmentation task provides region information to guide the boundary detection task's internal consistency modeling.
- (3) **Background Texture Interference.** Additionally, a multi-scale feature aggregation mechanism in the decoding stage enhances the network's ability to retain detailed features. The third situation is the complex texture of the background region can interfere with the model's recognition of lesion regions, leading to parts of the background being mistakenly identified as lesions. To address this, the feature selection capability of the proposed CCE module adaptively suppresses redundant background information, improving segmentation accuracy.

Despite achieving competitive performance, several challenges remain that warrant further investigation:

- (1) **Insufficient generalization ability across datasets:** The robustness of the model decreases when the dataset distribution significantly differs from that of the unlabeled data in the training set. To address this, future research should focus on developing stronger domain adaptation mechanisms to enhance the model's generalization ability across diverse datasets.
- (2) **Challenges with Complex Boundary Segmentation:** While the boundary detection task improves the segmentation task's perception of boundary regions, the model's performance remains suboptimal for particularly complex or highly irregular boundary regions in certain test samples. Future efforts could explore refined boundary optimization modules or incorporate shape-prior modeling to further enhance boundary segmentation capabilities.
- (3) **Exploration of Multimodal Data:** With the continuous advancement of medical imaging technology and the growing recognition of multimodal data's importance in medical image analysis, future work will focus on developing segmentation techniques for multimodal data.

This approach aims to provide more comprehensive and accurate insights for clinical applications.

5 Conclusions and future work

Considering the challenges of scarce labeled data, insufficient multi-scale feature fusion, and limited modeling of long-range dependencies in pneumonia CT segmentation, this paper proposes DTCR-U-Net, a novel lesion segmentation framework based on dual-task consistency regularization and MSAA. By jointly optimizing segmentation and boundary-detection branches, DTCR-U-Net employs a task-level consistency regularization loss to establish a cooperative optimization mechanism, effectively enhancing segmentation accuracy and boundary clarity. With MSAA, the designed CCE module optimizes feature weight distribution by capturing global dependencies across channels, while the MSGS module strengthens the focus on complex lesion regions by modeling cross-scale spatial dependencies. Furthermore, the MsSC mechanism facilitates efficient aggregation of multi-level features during decoding, ensuring robust performance across diverse datasets. Extensive experiments on multiple public benchmarks demonstrate that DTCR-U-Net consistently outperforms existing methods, particularly in delineating intricate boundaries and fine-grained lesion details. Its ability to leverage both labeled and unlabeled data in a semi-supervised manner further underscores its practical value for real-world clinical deployment.

Despite these promising results and the notable efficiency of our framework, several limitations remain and suggest directions for future work.

First, the core of our semi-supervised framework relies on the synergistic consistency between the segmentation and boundary detection tasks. While highly effective, its performance could be constrained in scenarios where the lesion boundaries are exceptionally ill-defined or discontinuous, as this may introduce noise into the consistency regularization signal. Future work could explore incorporating stronger topological priors or shape constraints to enhance robustness in such extreme cases.

Second, while the generalizability of DTCR-U-Net has been validated across multiple datasets, it still assumes that unlabeled and labeled data are drawn from reasonably similar distributions. Under severe domain shifts—for example, CT scans acquired with markedly different reconstruction kernels, noise characteristics, or imaging protocols—performance may degrade. Developing more advanced domain generalization strategies, beyond conventional domain adaptation, will be crucial to ensure reliable deployment across diverse clinical sites.

Finally, the current work focuses exclusively on a single imaging modality (CT), whereas many clinical decisions are made by integrating information from multiple sources. Extending DTCR-U-Net to exploit multimodal imaging (e.g., PET-CT, MRI-CT) and even non-imaging data (e.g., clinical reports, laboratory tests) represents an important and challenging direction for building more comprehensive and clinically informative decision-support systems.

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Declarations

Ethical approval Regarding the ethics statement requirement, our research does not involve experiments with human subjects. The study utilizes publicly available datasets, which have already obtained the necessary ethical approvals from their respective originating institutions. Since we did not collect any new data from human subjects and solely relied on pre-existing, ethically cleared public data for model development and validation, no additional ethical approval was required for this work.

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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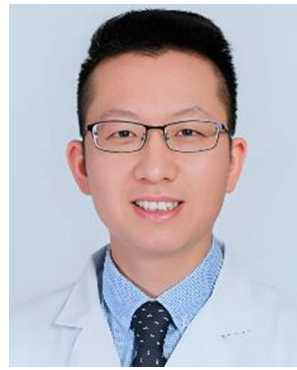
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